

SmartPark: Intelligent Vehicle Parking Management System

A Mini Real-World Java Application

Team Members:

Ann Mariya Sajeev -24UBC112

Vaishnavi Vijayan -24UBC158

Course: BCA

Internal Assessment – Mini Real-World Java Application

Course Outcome: CO5 – Develop and execute Java programs implementing object-oriented principles, GUI components, exception handling, multithreading, and database connectivity using JDBC.

1.Introduction

In today's rapidly evolving digital era, automation plays a crucial role in improving efficiency and minimizing human errors. Traditional parking management systems often rely on manual record keeping and human supervision, leading to inefficiencies, incorrect billing, and poor slot monitoring.

SmartPark is a desktop-based Java application designed to automate and streamline parking operations in real-world environments such as shopping malls, hospitals, corporate offices, airports, and residential complexes. The system provides an interactive GUI that allows users to manage parking slots, calculate billing automatically, and maintain records efficiently.

The project integrates Object-Oriented Programming principles, exception handling, multithreading, and JDBC database connectivity, thereby demonstrating practical implementation of Course Outcome CO5.

2.Problem Statement

Manual parking systems suffer from:

Duplicate vehicle entries

Incorrect billing calculations

No proper data storage

Lack of real-time slot monitoring

Human errors in record management

There is a need for an automated system that ensures accuracy, reliability, and efficiency in managing parking operations.

3.Objectives

The main objectives of SmartPark are:

To design a real-world Java GUI application

To implement core OOP concepts

To integrate JDBC for database connectivity

To apply exception handling for robust performance

To use multithreading for background processing

To simulate an intelligent parking management system

4.OOP Concepts Implemented

Encapsulation

Database operations are encapsulated within the ParkingDAO class.

Abstraction

The ParkingOperations interface defines core parking functionalities.

Polymorphism

Method overriding allows flexible booking and extension behavior.

Inheritance

The LiveUpdater class extends Thread to enable multithreading.

5.Key Features

Slot Booking

Enter vehicle number and owner name

Select number of hours

Automatic slot allocation

Duplicate vehicle detection

Slot color changes to Red

Extend Booking

Add extra hours

Automatic bill recalculation

Database update

Slot Reservation

Reserve slot in advance

Slot marked Blue

Prevents booking by others

Slot Release

Release occupied slot

Record removed from database

Slot becomes Available

Automated Billing System

₹50 per hour calculation

Instant total amount display

Dynamic billing logic

Payment Simulation

Simulated payment confirmation via:

GPay

PhonePe

Paytm

User-Friendly GUI

Developed using Java Swing:

Forms

Buttons

Dialog boxes

Color-coded slot grid

Real-time status display

6.Database Connectivity

Database Used: H2 Embedded Database

JDBC is used to:

Insert booking records

Update booking details

Delete released slots

Fetch parking data

Table Structure:

Field

Type

slot_id

INT

vehicle_no

VARCHAR

owner_name

VARCHAR

hours

INT

amount

DOUBLE

status

VARCHAR

7.Multithreading

The LiveUpdater thread:

Runs in the background

Refreshes slot status every 5 seconds

Keeps GUI responsive

Prevents application freezing

8.Exception Handling

Handled scenarios:

Empty input fields

Invalid vehicle number

Negative or zero hours

Duplicate vehicle booking

Database errors

Uses try-catch blocks and validation logic.

9. Technologies Used

Java

Java Swing

JDBC

H2 Database

Multithreading

Git & GitHub

10. Project Structure

Copy code

SmartPark/

|

|— src/

| |— SmartParkGUI.java

| |— ParkingDAO.java

| |— ParkingOperations.java

| |— LiveUpdater.java

|

```
├── lib/
|   └── h2.jar
|
├── screenshots/
|
└── README.md
```

11.Steps to Run

Clone the repository

Open in IntelliJ IDEA or NetBeans

Add H2 .jar file to libraries

Compile the project

Run SmartParkGUI.java

12.Sample Test Case

Input:

Vehicle No: KL07AB1234

Owner Name: Ammu

Hours: 3

Output:

Booking Successful

Total Amount: ₹150

Slot marked as Booked (Red)

13.Future Enhancements

Online payment gateway integration

MySQL cloud database

Admin dashboard

Report generation module

QR-based vehicle entry system

Mobile application version

14.Conclusion

SmartPark – Intelligent Vehicle Parking Management System successfully demonstrates the practical implementation of core Java programming concepts in a real-world scenario. The project integrates Object-Oriented Programming principles, GUI development using Java Swing, exception handling for robust execution, multithreading for responsive background processing, and JDBC-based database connectivity for persistent data management.

The system effectively automates parking operations such as slot booking, reservation, billing calculation, and record maintenance, thereby reducing human errors and improving operational efficiency. The use of a structured layered architecture ensures maintainability, scalability, and modularity of the application.

Through this project, theoretical knowledge has been transformed into a functional software solution that simulates real-time parking management environments like malls, hospitals, offices, and residential complexes. The implementation reflects strong understanding of software design principles, problem-solving skills, and application development practices.

Overall, SmartPark fulfills the requirements of Course Outcome CO5 and serves as a foundation for developing more advanced, enterprise-level management systems in the future.